



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 11 – 2011 YEARLY EXAMINATION

MATHEMATICS

ASSESSMENT TASK 4 WEIGHTING 40%

Monday 22nd August 2011
8.45 a.m. – 11.15 a.m.

OUTCOMES REFERRED TO: P2, P3, P4, P5, P6, P7, P8.

General Instructions

- Reading time – 5 minutes.
- Working time – $2\frac{1}{2}$ hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question.
- **Begin** each question on a **new page**.
- Write your **name**, **class** and your **teacher's name** on the front of each answer sheet of paper.

Total marks – 100

- Attempt Questions 1-10.
- All questions are of equal value.
- Mark values are shown at the side of each question part.

Question 1. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

- (a) Evaluate $\frac{\sqrt{\pi^2 + 9}}{2.1}$ correct to:

(i) 2 decimal places.

1

(ii) 2 significant figures.

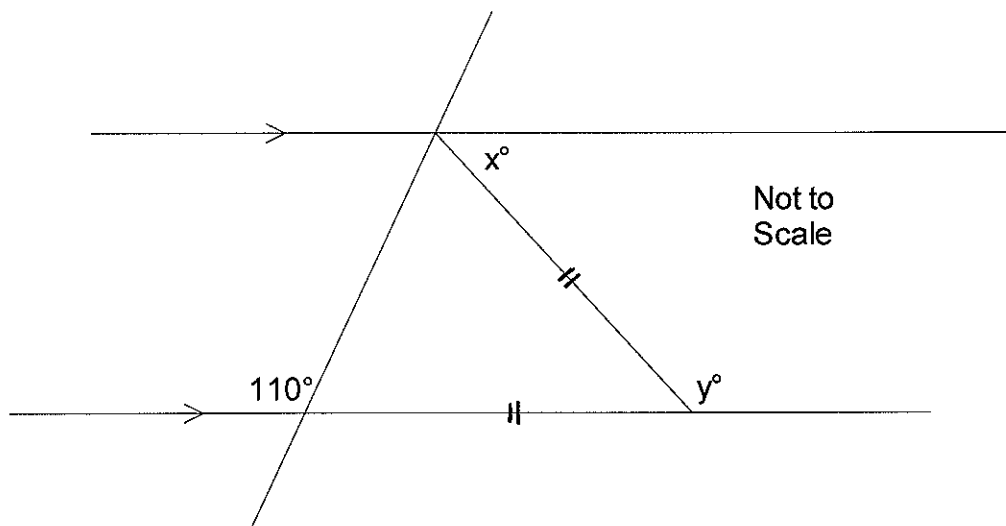
1

- (b) Factorise $a^2 + 13a - 30$.

2

- (c) Find the values of x and y .

2



- (d) The marshal paid \$139.50 for a meal at a restaurant. This included a 12.5% tip. What was the cost of the meal without the tip?

2

- (e) Expand and simplify $(2\sqrt{3} - 1)(\sqrt{3} + 4)$.

2

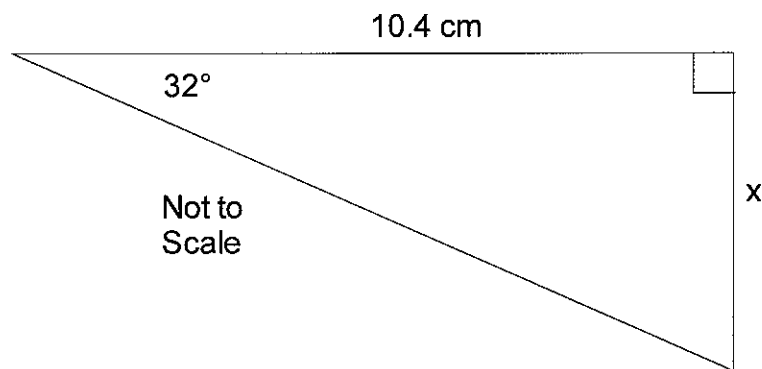
Question 2. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

- (a) State the equation of the horizontal line passing through $(-5, 1)$. 1

- (b) Solve $|2x + 1| \leq 7$. 2

- (c) Find the value of x correct to 1 decimal place. 2



- (d) Solve the following pair of simultaneous equations. 3
- $$3x - 2y = 10$$
- $$y = 2 - 2x$$

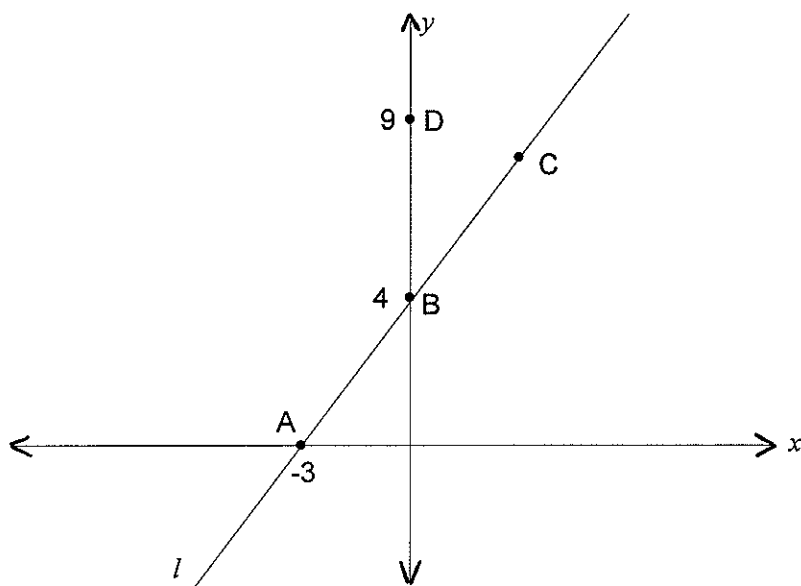
- (e) Find the integers a and b such that $\frac{1}{2 - \sqrt{3}} = a + b\sqrt{3}$. 2

Question 3. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

In the diagram below the line l cuts the x -axis at $A(-3, 0)$ and the y -axis at $B(0, 4)$.

The point D has the coordinates $(0, 9)$.



- | | | |
|-----|---|---|
| (a) | Find the gradient of the line l . | 1 |
| (b) | Hence, show that the equation of the line l is $4x - 3y + 12 = 0$. | 1 |
| (c) | The point C lies on l and B is the midpoint of the interval AC .
Find the coordinates of C . | 2 |
| (d) | Find the length of AD . | 2 |
| (e) | Show that AD is perpendicular to DC . | 2 |
| (f) | Find the equation of the line parallel to DC that passes through A . | 2 |

Question 4. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) Find the exact value of $\sec 30^\circ$. 1

(b) State the domain of the function $y = \sqrt{x-8}$. 1

(c) State the range of the function $y = |x| + 2$. 1

(d) Solve $\frac{x-5}{x+2} = \frac{3}{2}$. 2

(e) Simplify $\frac{x^3 - 27}{x^2 - 9}$. 2

(f) Indicate on a diagram using shading, the region for which the following inequalities hold simultaneously:

$$y \leq 1 - x^2 \quad \text{and} \quad y > x - 1. \quad \text{3}$$

Question 5. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) Solve the equation $\tan \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. **2**

(b) Differentiate with respect to x :

(i) $5x^3 + 2x - 6$. **1**

(ii) $\frac{6x-1}{3x+2}$. **2**

(iii) $4\sqrt{x} - \frac{1}{x^2}$. **2**

(c) Differentiate from first principles $f(x) = 4x^2 - 3x$. **3**

Question 6. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) If α and β are the roots of the equation $2x^2 + x - 4 = 0$, find the value of:

(i) $\alpha\beta + \alpha + \beta$. **2**

(ii) $\frac{1}{\alpha} + \frac{1}{\beta}$. **2**

(iii) $\alpha^2 + \beta^2$. **2**

(b) Simplify $\sqrt{1 - \cos^2 \theta}$. **1**

(c) Sketch on separate diagrams, the graphs of each of the following:

(i) $y = (x + 2)^2$. **1**

(ii) $x^2 + (y - 2)^2 = 4$. **2**

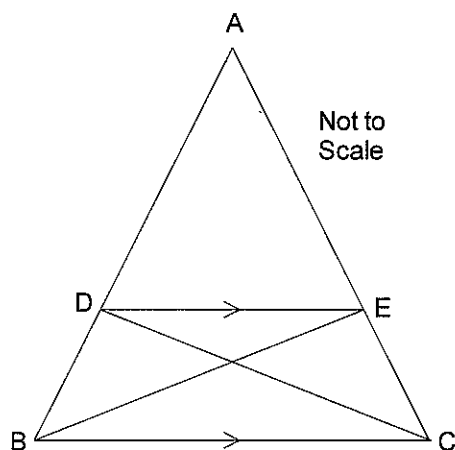
Question 7. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) Evaluate $\lim_{x \rightarrow -1} \frac{4x + 4}{x^2 + 5x + 4}$.

2

- (b) In the diagram, ABC is an isosceles triangle where $AB = AC$ and DE is parallel to BC.



Copy or trace your diagram on to your paper.

- (i) Show that ADE is an isosceles triangle.

2

- (ii) Show that $DB = EC$.

2

- (iii) Show that triangles DBC and ECB are congruent.

2

(c) Factorise fully $12e^2 - 14e - 6$.

2

Question 8. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

(a) Show whether the function $f(x) = x^2 + x$ is odd, even or neither. 2

(b) Find the values of k for which the expression $kx^2 - 2kx + 3$ is positive definite. 2

(c) Sketch the function $f(x) = \begin{cases} 2 & \text{if } x < -2 \\ -x & \text{if } -2 \leq x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$. 3

(d) Express the following with a rational denominator in its simplest form.

$$\frac{3\sqrt{2} + 2\sqrt{3}}{3\sqrt{2} - 2\sqrt{3}} \quad \text{3}$$

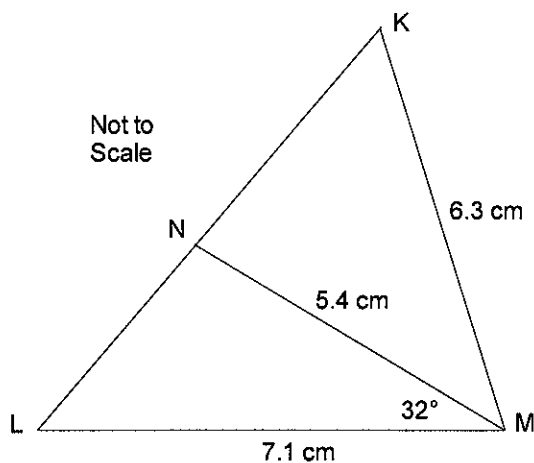
Question 9. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

- (a) Sketch the graph of $y = \cos x + 1$ for $0^\circ \leq x \leq 360^\circ$. 2

- (b) Prove $(\sec^2 \theta - 1) \cos^2 \theta = \sin^2 \theta$. 2

- (c) In the figure below, $KM = 6.3$ cm, $LM = 7.1$ cm, $NM = 5.4$ cm and $\angle NML = 32^\circ$.



Copy or trace the diagram on to your paper.

- (i) Find the size of $\angle LNM$. 2
- (ii) Find the length of NK. 3
- (iii) Find the area of $\triangle KLM$. 1

Question 10. (10 marks) Start each question a new page.

Write your name and your teacher's name at the top of each page.

- (a) Solve the following pair of simultaneous equations.

$$4a + 3b = 3$$

$$3a - 2b = -19$$

3

- (b) Solve $\left(x + \frac{1}{x}\right)^2 - 6\left(x + \frac{1}{x}\right) + 8 = 0.$

3

- (c) Find the equation of the normal to the curve $y = (2x + 1)^2$
at the point where $x = -1.$

4

End of paper

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Q1.

2.068531329

a) (i) 2.07 (1)

(ii) 2.1 (1)

b) $a^2 + 13a - 30$
 $= (a + 15)(a - 2)$
(1) (1)

c) $x = 40^\circ$ (1)
 $y = 140^\circ$ (1)

d) $1.125x = 139.50$
 $x = \frac{139.50}{1.125}$ (M1)
 $= \$124$ (1)

e) $(2\sqrt{3} - 1)(\sqrt{3} + 4)$
 $= 2(3) + 8\sqrt{3} - \sqrt{3} - 4$ (M1)
 $= 2 + 7\sqrt{3}$ (1)

Q2.

a) $y = 1$ (1)

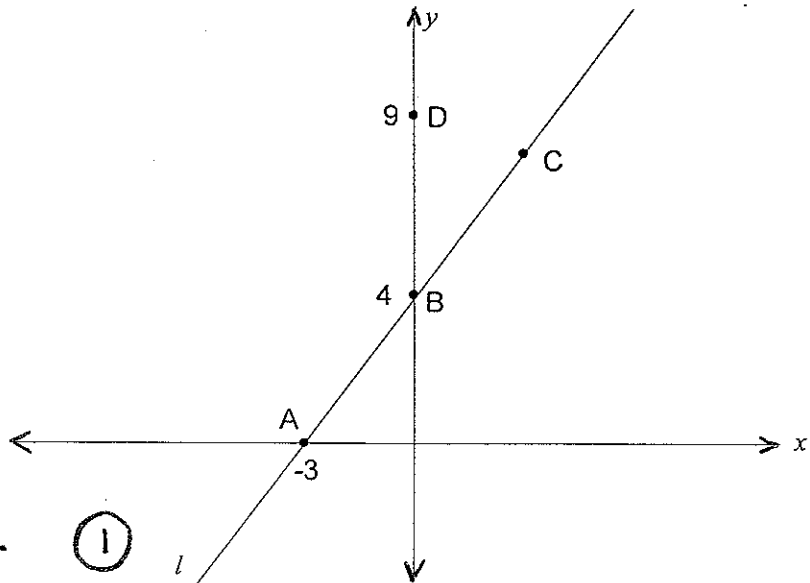
b) $-7 \leq 2x + 1 \leq 7$ (M1)
 $-8 \leq 2x \leq 6$
 $-4 \leq x \leq 3$ (1)

c) $\tan 32^\circ = \frac{x}{10.4}$ (M1)
 $x = 10.4 \times \tan 32^\circ$
 $= 6.5 \text{ cm (1 d.p.)}$ (1)

d) $3x - 2(2 - 2x) = 10$ (M1)
 $3x - 4 + 4x = 10$
 $7x = 14$
 $x = 2$ (1)
 $y = 2 - 2(2)$
 $y = -2$ (1)

e) $\frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}}$
 $= \frac{2 + \sqrt{3}}{4 - 3}$
 $= 2 + \sqrt{3}$ where $a = 2$ and $b = 1$
(1) (1)

Q3.



a) $m_{AC} = \frac{4}{3}$ (1)

b) $y - 4 = \frac{4}{3}(x - 0)$
 $3y - 12 = 4x$ (M1)
 $\therefore 4x - 3y + 12 = 0$

c) $\frac{x + (-3)}{2} = 0$ $\frac{y + 0}{2} = 4$
 $\therefore x = 3$ (1) $y = 8$ (1)

d) $AD = \sqrt{(-3-0)^2 + (0-9)^2}$ (M1)
 $= \sqrt{9 + 81}$
 $= \sqrt{90}$ OR $3\sqrt{10}$ (1)

e) $m_{AD} = \frac{9}{3} = 3$ $m_{DC} = \frac{8-9}{3-0} = -\frac{1}{3}$ (M1)
 $AD \perp DC$ since $3 \times (-\frac{1}{3}) = -1$ (1)

f) $y - 0 = -\frac{1}{3}(x - (-3))$ (M1)
 $y - 0 = -\frac{1}{3}(x + 3)$
 $y = -\frac{1}{3}x - 1$ (1)

Q4.

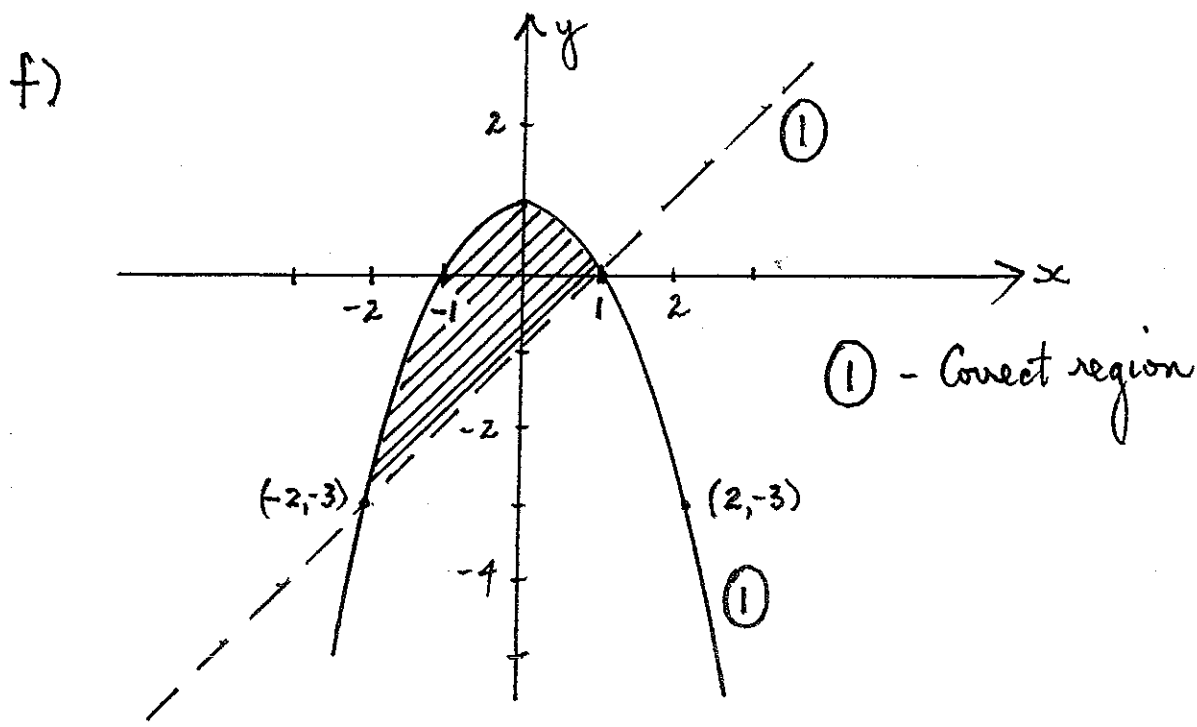
a) $\sec 30^\circ = \frac{2}{\sqrt{3}}$ OR $\frac{2\sqrt{3}}{3}$ (1)

b) Domain: $x \in \mathbb{R}, x \geq 8$ (1)

c) Range: $y \in \mathbb{R}, y \geq 2$ (1)

d) $\frac{x-5}{x+2} = \frac{3}{2}$
 $2x-10 = 3x+6$ (M1)
 $-16 = x$
 $x = -16$ (1)

e) $\frac{(x-3)(x^2+3x+9)}{(x-3)(x+3)}$ (M1)
 $= \frac{x^2+3x+9}{x+3}$ (1)



Q5

a) $\tan \theta = -1$

$\therefore \theta = 135^\circ, 315^\circ$
① ①

b) i) $\frac{d}{dx}(5x^3 + 2x - 6)$
 $= 15x^2 + 2$ ①

ii) $y' = \frac{(3x+2)(6) - (6x-1)(3)}{(3x+2)^2}$ (M1)
 $= \frac{18x + 12 - 18x + 3}{(3x+2)^2}$
 $= \frac{15}{(3x+2)^2}$ ①

iii) $f(x) = 4x^{\frac{1}{2}} - x^{-2}$
 $f'(x) = 2x^{-\frac{1}{2}} + 2x^{-3}$
① ①

c) $f(x) = 4x^2 - 3x$
 $f(x+h) = 4(x^2 + 2xh + h^2) - 3(x+h)$
 $= 4x^2 + 8xh + 4h^2 - 3x - 3h$

$f'(x) = \lim_{h \rightarrow 0} \frac{4x^2 + 8xh + 4h^2 - 3x - 3h - (4x^2 - 3x)}{h}$ (M1)
 $= \lim_{h \rightarrow 0} \frac{\cancel{4x^2} + 8xh + 4h^2 - \cancel{3x} - 3h - \cancel{4x^2} + \cancel{3x}}{h}$
 $= \lim_{h \rightarrow 0} \frac{\cancel{h}(8x + 4h - 3)}{\cancel{h}}$ (M1)
 $= \lim_{h \rightarrow 0} 8x + 4h - 3$
 $= 8x - 3$ ①

Q6

$$a) \quad \alpha + \beta = -\frac{1}{2} \quad \alpha\beta = -\frac{4}{2} = -2$$

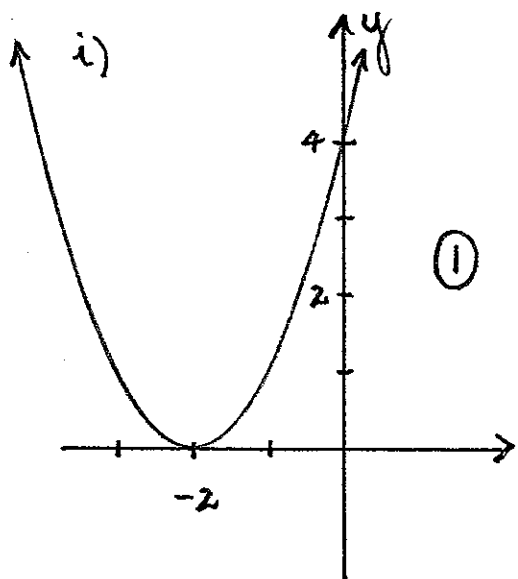
$$i) \quad \alpha\beta + (\alpha + \beta) \\ = -2 - \frac{1}{2} \quad (M1) \\ = -2\frac{1}{2} \quad (1)$$

$$ii) \quad \frac{1}{\alpha} + \frac{1}{\beta} \\ = \frac{\beta + \alpha}{\alpha\beta} \quad (M1) \\ = \frac{-\frac{1}{2}}{-2} \\ = \frac{1}{4} \quad (1)$$

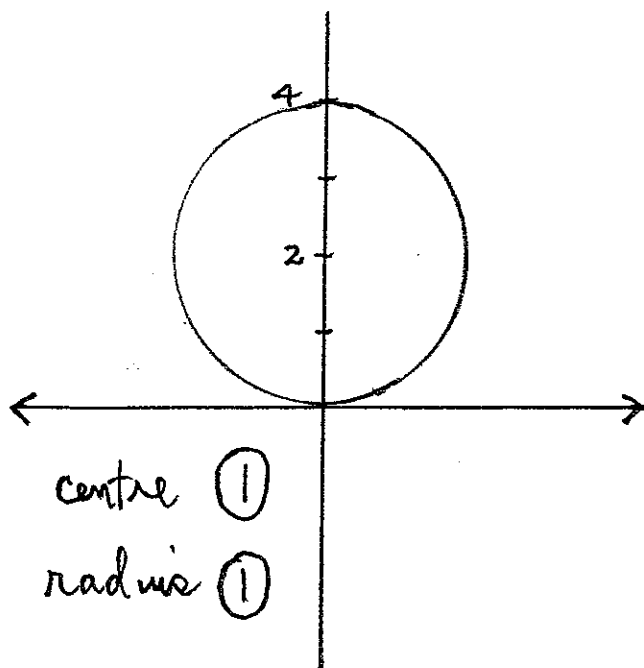
$$iii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \\ = \left(-\frac{1}{2}\right)^2 - 2(-2) \quad (M1) \\ = 4\frac{1}{4} \quad (1)$$

$$b) \quad \sqrt{1 - \cos^2 \theta} \\ = \sqrt{\sin^2 \theta} \\ = \sin \theta \quad (1)$$

c)

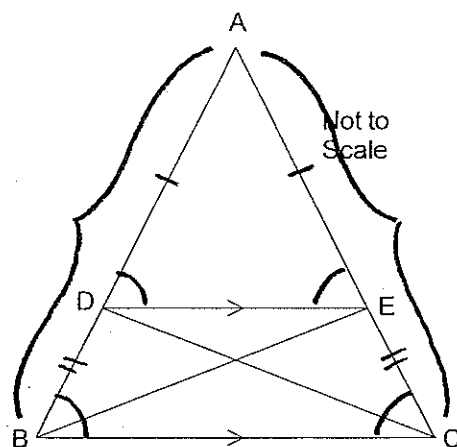


ii)



Q7. a) $\lim_{x \rightarrow -1} \frac{4(\cancel{x+1})}{(\cancel{x+1})(x+4)} \quad (M1)$
 $= \lim_{x \rightarrow -1} \frac{4}{x+4}$
 $= \frac{4}{3} \quad (1)$

b)



(Absolutely important to indicate 'equality' and/or direct reference to parallel lines.)

i) $\angle ABC = \angle ACB$ ΔABC is isosceles
 $\angle ADE = \angle ABC$ } Corr. L's = . ($DE \parallel BC$) (1)
 $\angle AED = \angle ACB$
 $\therefore \angle ADE = \angle AED$ Equal L's are equal (1)
 $\therefore \Delta ADE$ is isosceles

any other valid approach but must use correct reasons that apply to that method.

ii) $AB = AC$ Given (1)
 $AD = AE$ ΔADE is isosceles (1)
 $AB - AD = AC - AE$ Equals subtracted from (1)
 $\therefore DB = EC$ equals are =.
 iii) $DB = EC$ Proven above
 $\angle ABC = \angle ACB$
 $BC = BC$ BC is a common side (1)
 $\Delta DBC \equiv \Delta ECB$ S.A.S (1)

c) $2(6e^2 - 7e - 3) \quad (1)$
 $= 2(3e + 1)(2e - 3) \quad (1)$

(must be completely factorised for last mark)

Q8.

a)

$$f(x) = x^2 + x$$

$$f(-x) = (-x)^2 + (-x) \quad (M1)$$

$$= x^2 - x$$

$$f(-x) \neq f(x)$$

$$f(-x) \neq -f(x) \therefore f(x) \text{ is neither odd or even} \quad (1)$$

b)

$$\Delta < 0 \text{ and } k > 0 \quad (M1)$$

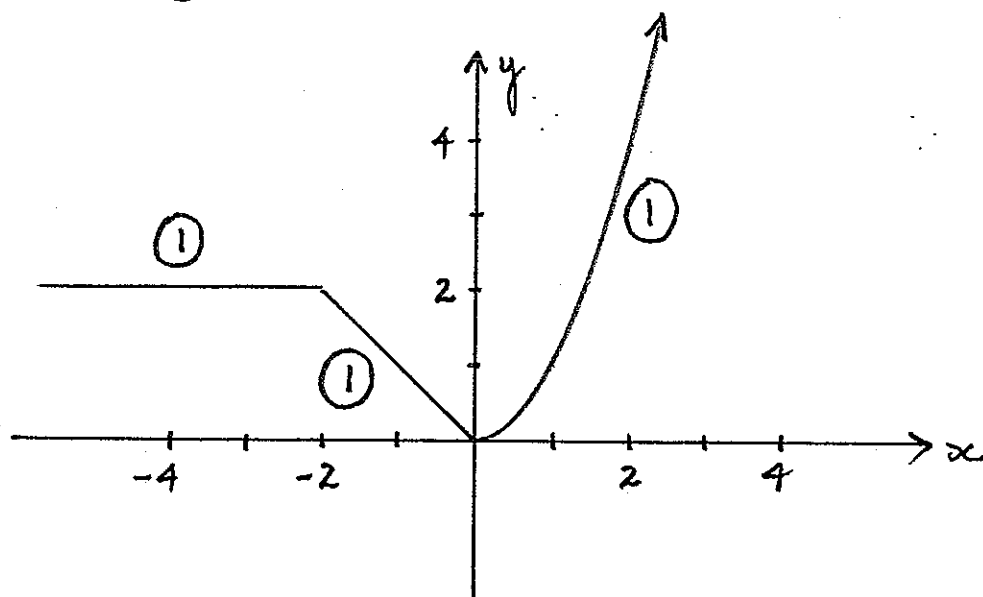
$$(-2k)^2 - 4(k)(3) < 0$$

$$4k^2 - 12k < 0$$

$$4k(k-3) < 0$$

$$0 < k < 3 \quad (1)$$

c)



d)

$$\frac{3\sqrt{2} + 2\sqrt{3}}{3\sqrt{2} - 2\sqrt{3}} \times \frac{3\sqrt{2} + 2\sqrt{3}}{3\sqrt{2} + 2\sqrt{3}}$$

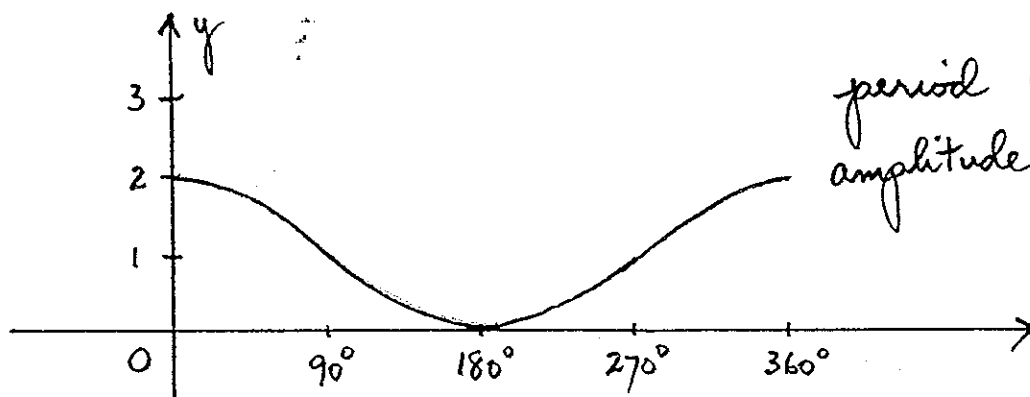
$$= \frac{18 + 12\sqrt{6} + 12}{18 - 12} \quad (M1)$$

$$= \frac{30 + 12\sqrt{6}}{6} \quad (M1)$$

$$= 5 + 2\sqrt{6} \quad (1)$$

Q9.

a)

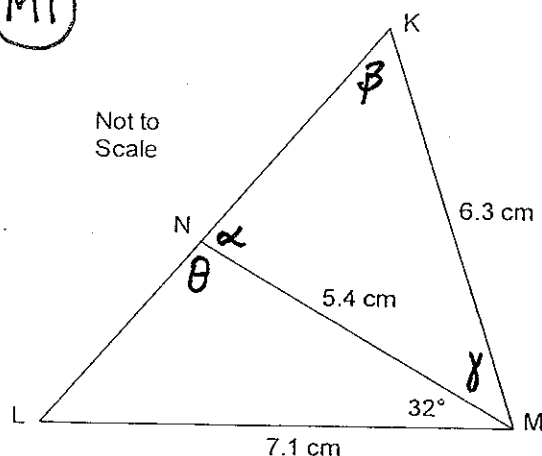


period ①
amplitude ①

$$\begin{aligned} b) \quad & (\sec^2 \theta - 1) \cos^2 \theta \\ &= \tan^2 \theta \cos^2 \theta \quad (M1) \\ &= \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos^2 \theta \\ &= \sin^2 \theta \quad (1) \end{aligned}$$

c) i)

$$\begin{aligned} LN^2 &= 5.4^2 + 7.1^2 - 2(5.4)(7.1) \cos 32^\circ \\ &= 14.54167199 \\ LN &= 3.8 \text{ cm (1 d.p.)} \quad (M1) \end{aligned}$$



$$\begin{aligned} \text{ii) } \frac{\sin \angle LNM}{7.1} &= \frac{\sin 32^\circ}{3.813354427} \\ \angle LNM &= \sin^{-1} \left(\frac{7.1 \times \sin 32^\circ}{3.813354427} \right) \\ \theta &= 80^\circ 38' \text{ OR } 80.6^\circ \quad (1) \end{aligned}$$

$$\begin{aligned} \text{iii) } \therefore \alpha &= 180 - 80.6 \\ &= 99.4^\circ \quad (1) \end{aligned}$$

$$\frac{\sin 99.4}{6.3} = \frac{\sin \angle NKM}{5.4}$$

$$\begin{aligned} \beta &= \sin^{-1} \left[\frac{5.4(\sin 99.4)}{6.3} \right] \\ &= 57.7^\circ \quad (1) \end{aligned}$$

$$\begin{aligned} \angle NMK &= 180 - (57.7 + 99.4) \\ \gamma &= 22.9^\circ \end{aligned}$$

$$\begin{aligned} \frac{NK}{\sin 22.9^\circ} &= \frac{6.3}{\sin 99.4^\circ} \\ NK &= \frac{6.3(\sin 22.9^\circ)}{\sin 99.4^\circ} = 2.5 \text{ cm} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{iii) } A &= \frac{1}{2}(7.1)(6.3) \sin(32^\circ + 22.9^\circ) \\ &= 18.3 \text{ cm}^2 \quad (1) \end{aligned}$$

$$\begin{aligned} \cos \theta &= \frac{3.8133...^2 + 5.4^2 - 7.1^2}{2(3.8133...)(5.4)} \\ &= -0.1628858514 \\ \therefore \theta &= 99^\circ 22' \text{ OR } 99.4^\circ \quad (1) \end{aligned}$$

$$\begin{aligned} \text{iii) } \therefore \alpha &= 180 - 99.4 \\ &= 80.6^\circ \end{aligned}$$

$$\frac{\sin 80.6}{6.3} = \frac{\sin \angle NKM}{5.4}$$

$$\begin{aligned} \beta &= \sin^{-1} \left[\frac{5.4(\sin 80.6^\circ)}{6.3} \right] \\ &= 57.7^\circ \quad (1) \end{aligned}$$

$$\begin{aligned} \angle NMK &= 180 - (80.6 + 57.7) \\ \gamma &= 41.7^\circ \end{aligned}$$

$$\begin{aligned} \frac{NK}{\sin 41.7^\circ} &= \frac{6.3}{\sin 80.6^\circ} \\ NK &= \frac{6.3(\sin 41.7^\circ)}{\sin 80.6^\circ} = 4.2 \text{ cm} \quad (1) \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2}(7.1)(6.3) \sin(32^\circ + 41.7^\circ) \\ &= 21.5 \text{ cm}^2 \quad (1) \end{aligned}$$

Q10.

$$\begin{array}{rcl}
 a) & 4a + 3b = 3 & \dots \textcircled{1} \\
 & 3a - 2b = -19 & \dots \textcircled{2} \\
 & 8a + 6b = 6 & \textcircled{M1} \\
 & 9a - 6b = -57 & \textcircled{M1} \\
 \hline
 & 17a = -51 & \\
 & a = -3 & \textcircled{1}
 \end{array}$$

$$\begin{array}{rcl}
 4(-3) + 3b = 3 \\
 3b = 15 \\
 b = 5 & \textcircled{1}
 \end{array}$$

$$\begin{array}{l}
 b) \text{ let } w = x + \frac{1}{x} \\
 w^2 - 6w + 8 = 0 \quad \textcircled{M1} \\
 (w - 4)(w - 2) = 0 \\
 x + \frac{1}{x} - 4 = 0 \quad x + \frac{1}{x} - 2 = 0 \\
 x^2 - 4x + 1 = 0 \quad x^2 - 2x + 1 = 0 \\
 x = \frac{-(-4) \pm \sqrt{16 - 4(1)(1)}}{2(1)} \quad (x - 1)(x - 1) = 0 \\
 = \frac{4 \pm 2\sqrt{3}}{2} \\
 = 2 \pm \sqrt{3} \quad \textcircled{1} \\
 x = 1 \quad \textcircled{1}
 \end{array}$$

$$\begin{array}{l}
 c) \quad y = (2x + 1)^2 \\
 \frac{dy}{dx} = 4(2x + 1) \quad \textcircled{M1} \\
 \text{when } x = -1, \quad y = (-2 + 1)^2 = 1 \quad (-1, 1) \quad \textcircled{1}
 \end{array}$$

$$\begin{array}{l}
 m = \frac{dy}{dx} = 4(-1) = -4 \\
 \perp m = \frac{1}{4} \quad \textcircled{1} \\
 y - 1 = \frac{1}{4}(x + 1) \\
 y = \frac{1}{4}x + \frac{5}{4} \quad \text{OR} \quad x - 4y + 5 = 0 \quad \textcircled{1}
 \end{array}$$